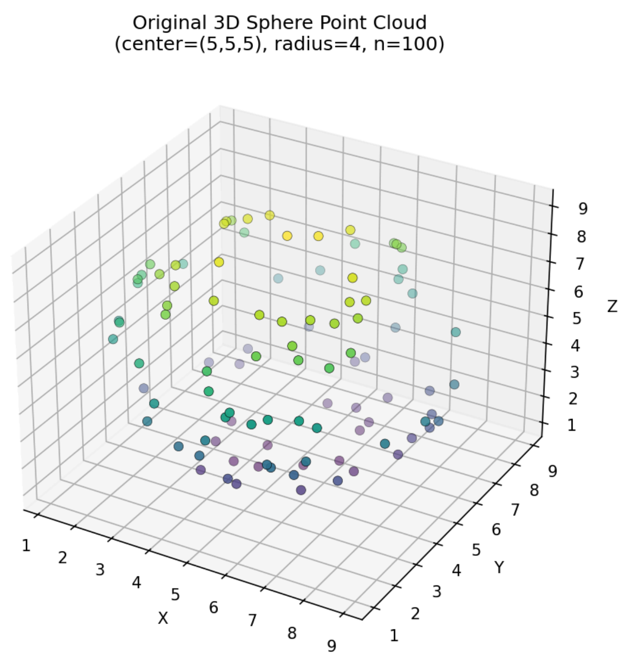
Deep Learning Lab Assignment - 1

Methodology:

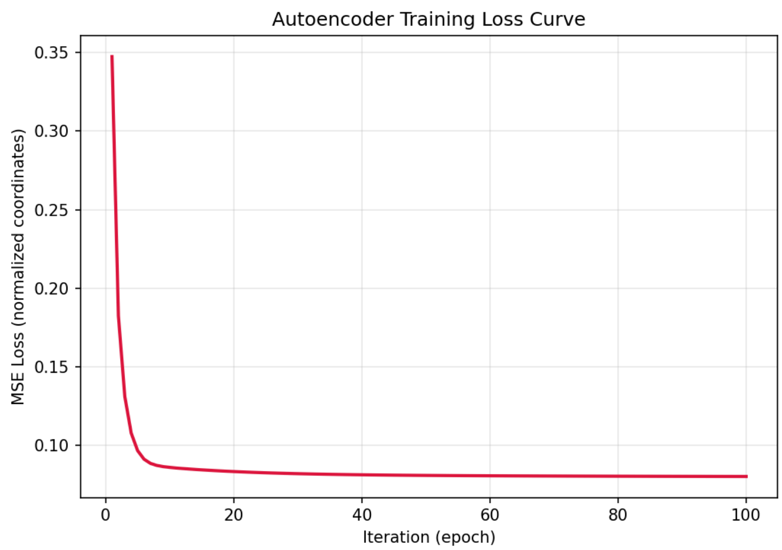
A total of 100 uniformly distributed points were generated on the surface of a sphere with center (5, 5, 5) and radius 4 by creating random 3D vectors, normalizing them, scaling by the radius, and translating them to the sphere's center. The coordinates were then normalized by subtracting the center and dividing by the radius to improve training stability. A 3 → 2 → 3 Autoencoder was built, where the encoder compresses the 3D coordinates into a 2-dimensional latent representation using a tanh activation function, and the decoder reconstructs the original coordinates through a linear output layer. The network was trained for 100 epochs using full-batch gradient descent (batch size = 100) with Mean Squared Error (MSE) as the loss function. After training, the encoder was used to extract the 2D latent representations, and the model's performance was evaluated using the final reconstruction MSE.

Results

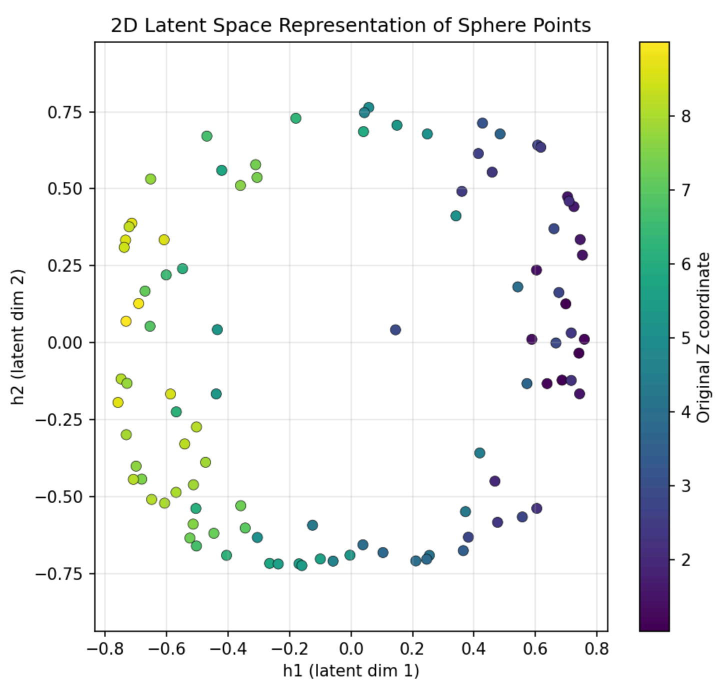
- Original 3D sphere point cloud:



- Training loss curve:



- 2D latent space representation:



- Summary of results:

Metric	Value
Epochs (iterations)	100
Batch size	100 (full batch)
Architecture	3 -> 2 (bottleneck) -> 3
Hidden activation	tanh
Output activation	linear
Loss at epoch 1	0.347461
Loss at epoch 100 (normalized coords)	0.080004
Final reconstruction MSE (original coord. scale)	1.279953 units^2

Experimental Analysis

1. Why is an Autoencoder considered an unsupervised learning model although the input and output are identical during training?

Ans: An Autoencoder is an unsupervised learning model because it does not require labeled data. It learns by reconstructing the input itself, discovering meaningful patterns and features without external supervision.

2. What is the significance of the bottleneck layer in an Autoencoder?

The bottleneck layer compresses the input into a lower-dimensional representation. It forces the network to retain only the most important features, enabling dimensionality reduction and efficient feature extraction.

3. Why is it possible to represent a 3D sphere using only two latent variables?

Although a sphere is represented in 3D coordinates, its surface is intrinsically 2-dimensional. Every point on the sphere can be described using two parameters (e.g., latitude and longitude), so two latent variables are sufficient.

4. How would the reconstruction error change if the bottleneck size were increased to 3 or reduced to 1? Justify your answer.

Ans:
Bottleneck = 3: Reconstruction error decreases because the model has more capacity and can nearly learn the identity mapping.
Bottleneck = 1: Reconstruction error increases since one latent variable cannot capture the 2D structure of the sphere, causing information loss.

5. Discuss whether the learned latent space preserves the neighbourhood relationships among the original 3D points.

Yes, the latent space generally preserves local neighbourhood relationships, meaning nearby points on the sphere remain close in the latent space. However, some global distortions may occur because a curved 3D surface is being represented in a 2D space.